Checkout lines: probability

CHEAT SHEET

@nova.xplains

What is probability?



Probability = a number between 0 (never) and 1 (always)

0.5 = 1/2 = 50 %: maybe, like heads or tails

Coin: 1/2 for heads · Die: 1/6 for a six

The Laplace rule

Laplace rule

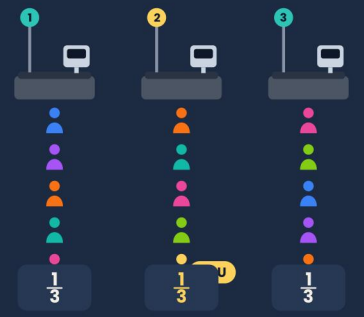
$$P = \frac{\text{results you want}}{\text{all possible results}}$$

Probability = wanted outcomes ÷ all possible outcomes

Only works if every outcome is equally likely (fair)

Example: even number on a die = 3/6 = 1/2

Three checkouts



Three lines, same chance: yours is fastest only 1/3 of the time

Complement: 1 - 1/3 = 2/3, so another line wins twice as often

With 5 checkouts: yours wins 1/5, another one 4/5

Many shopping trips

$$\frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3}$$

$$\frac{1}{243}$$

almost impossible

Independent events: multiply the probabilities

Lucky twice: 1/3 × 1/3 = 1/9

Lucky 5 trips in a row: (1/3)^5 = 1/243

Your brain



Wins are forgotten, losses stick (negativity bias)

Waiting feels about 36 % longer than it really is

So it FEELS like you always pick the wrong line

The fairest line



One shared line for all checkouts = queueing theory (A. K. Erlang, 1909)

A slow customer only blocks one checkout, the line keeps moving

Same idea: traffic lanes, elevators, call centers

BONUS It is not bad luck, it is math! Quiz: with 4 checkouts, your line is the fastest 1 time in 4, another one 3 times in 4.

Sources: Bill Hammack (engineerguy.com), Alex Stone (The New York Times, 2012), Richard Larson (MIT), A. K. Erlang (1909), Laplace probability

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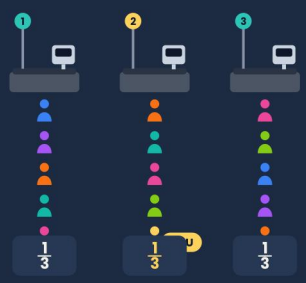
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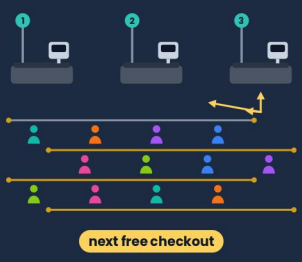


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